Grade 7 Accelerated
Unit 3
Lesson 6 Practice

Name_____Answer Key____
Date____________________
Period__________________

Apply the laws of exponents to evaluate each expression. Show your steps in the process.

1. $\frac{(0.25)^{-2} \cdot (0.25)^{5}}{(0.25)^{3}}$

$\frac{(0.25)^{3}}{(0.25)^{3}} = 1$

2. $(-3)^{3} \cdot \left(\frac{-5}{3}\right)^{-4}$

$(-3)^{3} \cdot \left(\frac{-3}{5}\right)^{4} = \frac{(-3)^{7}}{5^{4}} = \frac{-2187}{625}$

For each expression, apply one or more exponent laws to evaluate.

Expression	**Value**
3. $(3^{3})^{-3}$	$3^{-9} = \frac{1}{3^{9}} = \frac{1}{19683}$
4. $\left(\frac{48}{6}\right)^{-2}$	$(\frac{6}{48})^{2} = \frac{36}{2304} = \frac{9}{576} = \frac{1}{64}$
5. $(3^{4} \cdot 2^{-4})^{-1}$	$3^{-4} \cdot 2^{4} = \frac{1}{3^{4}} \cdot 2^{4} = \frac{16}{81}$

Evaluate each expression in the table below. Then identify which exponent law(s) were used to evaluate each expression.

Expression	Value	Exponent Law(s)
6. $\frac{6^{-2} \cdot 6^{5}}{6}$	36	Answers May Vary
7. $\frac{-9^{4}}{-9^{0} \cdot -9^{3}}$	-9	Answers May Vary
8. $5^{2} \cdot \left(\frac{1}{4}\right)^{-1}$	100	Answers May Vary
9. $\frac{(-2.5)^{0} \cdot (-2.5)^{1}}{(-2.5)^{-2}}$	-15.625	Answers May Vary
10. $\left(\frac{7}{5}\right)^{-4}$	$\frac{625}{2401}$	Answers May Vary